5 During cosmic-ray experiments in the 1930s, researchers observed the track of a previously unknown positively charged particle in a cloud chamber. The particle was measured to have a charge of +1 *e* and is known as the muon.

The particle enters a uniform magnetic field of flux density 0.250 T with a speed of 2.40×10^{7} m s^{-1}. Its velocity makes an angle of 60° with the direction of the magnetic field, causing it to follow a helical path as seen in Fig. 5.1 below.

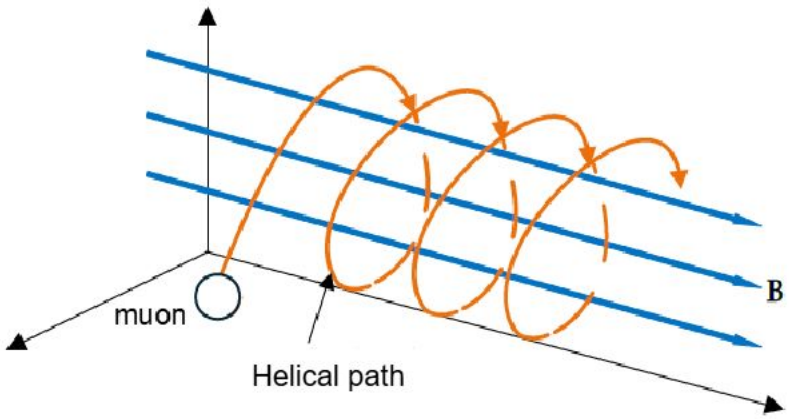


Fig. 5.1

Measurements of the track show that the muon took 2.96×10^{-8} s to complete one revolution.

(a) (i) Show that the radius of the helical path is 0.0979 m.

$$v_{\perp} = r\omega = r\frac{2\pi}{T}$$

$$r = \frac{v_{\perp}T}{2\pi}$$

$$= \frac{2.40\times10^{7}\sin60^{\circ}\times2.96\times10^{-8}}{2\pi}$$

$$= 0.09792$$

$$\approx 0.979 \text{ m}$$

[2]

(ii) Determine the mass, m_{μ} of the muon.

The magnetic force provides the centripetal force:

$$Bqv_{\perp} = \frac{mv_{\perp}^{2}}{r}$$ [1]

$$m = \frac{Bqr}{v_{\perp}}$$

$$= \frac{0.250\times1.60\times10^{-19}\times0.0979}{2.40\times10^{7}\sin60^{\circ}}$$

$$= 1.88\times10^{-28} \text{ m}$$ [1]

m_{μ} = ... kg [2]

(b) Determine the distance travelled by the particle **parallel** to the magnetic field during one complete revolution.

Distance moved in 1 revolution

$d = v_{\parallel} T$

$= v \cos 60^\circ T$ [1 for correct resolution]

$= 2.4 \times 10^7 \times \cos 60^\circ \times 2.96 \times 10^{-8}$

$= 0.355$ m [1]

distance = ... m [2]

(c) Before taking measurements, the researchers compensate for the vertical component of the Earth's magnetic field.

A short magnetic needle is suspended from a string 5.0 cm from a horizontal long straight wire, as shown in the top view in Fig. 5.2.

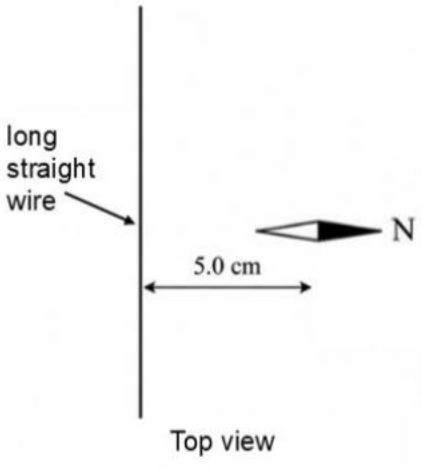


Fig. 5.2

When there is no current in the wire, the north pole of the needle dips at an angle of 6.0° below the horizontal, as shown in the side view in Fig. 5.3.

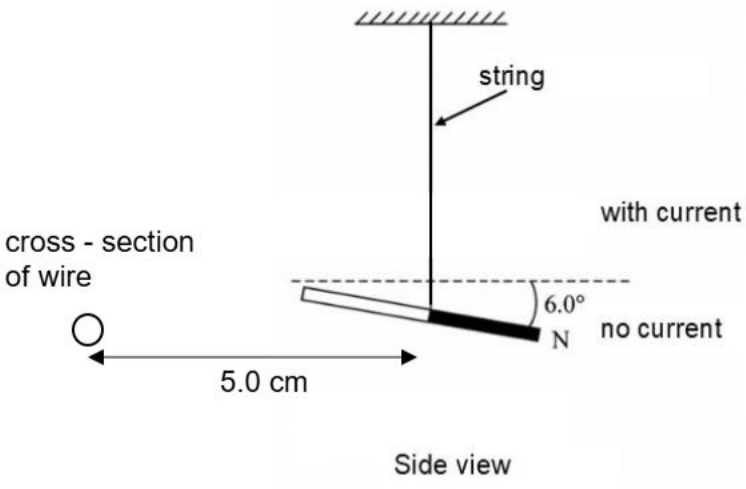


Fig. 5.3

A current of 0.48 A is then passed through the wire. The magnetic field produced by the current causes the needle to become horizontal, as shown in the side view in Fig. 5.4.

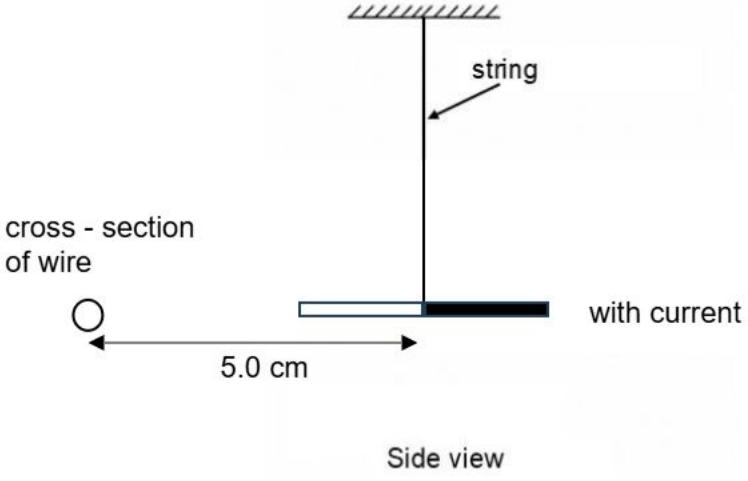


Fig. 5.4

(i) Show that the magnetic flux density produced by the wire, when the current is flowing through it, at the position of the short magnetic needle is 1.9×10^{-6} T.

The magnetic field due to a long straight wire is

$$B_{\text{wire}} = \frac{\mu_0 I}{2\pi r}$$

$$= \frac{(4\pi \times 10^{-7})(0.47)}{2\pi(0.050)}$$

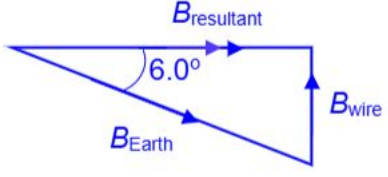
$$= 1.88 \times 10^{-6} \text{ T} \qquad [1]$$

$$\approx 1.9 \times 10^{-6} \text{ T} \qquad [1]$$

(ii) Determine the magnitude of the Earth's magnetic flux density in the laboratory.

The Earth's magnetic field makes an angle of 6.0° below the horizontal.
The wire's field is vertical at the position of the needle, so for the needle to become horizontal, the vertical component of the Earth's field must be cancelled.

$$B_{wire} = B_{Earth} \sin 6.0^\circ$$ [1]



$$B_{Earth} = \frac{1.88 \times 10^{-6}}{\sin 6.0^\circ}$$

$$= 1.80 \times 10^{-5} \text{ T}$$ [1]

magnetic flux density = ...T [2]

Total: [9]